

DP2: Control System Design for Flying Wing Landing

Maxwell Lutter * and Navid Navidzadeh *
University of Illinois at Urbana-Champaign, Urbana, Illinois, 61801

I. Nomenclature

$(\cdot)$	=	time derivative
A	=	state transition matrix
B	=	input matrix
F	=	closed-loop system matrix
$f(m, n)$	=	function for equations of motion
$g(x, u)$	=	running cost vector function
h	=	terminal cost function
K	=	LQR state-feedback gain matrix
M	=	terminal cost weight matrix
m	=	State vector
n	=	Input vector
P	=	solution to the algebraic Riccati equation
p_x, p_y, p_z	=	position in x, y, and z directions (m)
Q	=	state-weighting matrix (Positive Semidefinite)
R	=	input-weighting matrix (Positive Definite)
W	=	Controllability matrix
$v(t, x)$	=	value vector function (optimal cost-to-go)
v_x, v_y, v_z	=	velocity in x, y, and z directions (m/s)
w_x, w_y, w_z	=	roll rate, pitch rate, yaw rate (rad/s)
x	=	state vector with respect to equilibrium state
δ_r, δ_l	=	elevon angle input
ψ, θ, ϕ	=	roll, pitch, and yaw (rad)

II. Theory

A. Equations of motion

The goal of this project is to design a controller which shall guide a flying wing to land on an airstrip using elevon inputs. This system's equations of motion are given in standard form, provided by Professor Bretl and Professor Chang:

$$\dot{m} = f(m, n), \quad (1)$$

$$\begin{bmatrix} \dot{p}_y \\ \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \\ \dot{v}_x \\ \dot{v}_y \\ \dot{v}_z \\ \dot{w}_x \\ \dot{w}_y \\ \dot{w}_z \end{bmatrix} = f \left(\begin{bmatrix} p_y \\ \psi \\ \theta \\ \phi \\ v_x \\ v_y \\ v_z \\ w_x \\ w_y \\ w_z \end{bmatrix}, \begin{bmatrix} \delta_r \\ \delta_l \end{bmatrix} \right). \quad (2)$$

*Undergraduate Student, Aerospace Engineering

A full definition of these equations of motion is shown in the code. The x-position, p_x , and z-position, p_z , are omitted to make equilibrium possible, since some change in position in the x and z direction is necessary to remain airborne in unpowered flight. This is permissible because these variables do not appear in any other ODEs defining the system, so eliminating these states does not affect any others.

B. Determining Equilibrium

The equilibrium state is defined in terms of the equations of motion:

$$\vec{0} = \vec{f}(m_e, n_e) \quad (3)$$

The equations of motion are not able to be solved analytically because they include many nonlinear terms, so a numerical solution is found using Least Squares optimization from an initial guess. For a guess,

$$m_{\text{guess}} = \begin{bmatrix} 0.0 & 0.0 & 0.05 & 0.0 & 10. & 0.0 & -0.4 & 0.0 & 0.0 & 0.0 \end{bmatrix}^T, \quad (4)$$

$$n_{\text{guess}} = \begin{bmatrix} -0.1 & -0.1 \end{bmatrix}^T. \quad (5)$$

The least squares search finds equilibrium points by finding a local minimum of $|\vec{f}(m_e, n_e)|$:

$$m_e = \begin{bmatrix} 0.0 & 0.0 & 0.004 & 0.0 & 10.1153 & 0.0 & 0.746 & 0.0 & 0.0 & 0.0 \end{bmatrix}^T, \quad (6)$$

$$n_e = \begin{bmatrix} -0.1833 & -0.1833 \end{bmatrix}^T. \quad (7)$$

This guess was chosen by restricting any values which must be or likely would be zero to zero, and then adjusting the remaining parameters to find an equilibrium input within the bounds of the elevon range $[-0.5, 0.5]$. This was done by guess and check with some physical intuition. The linearized system can be expressed in state space form as:

$$\dot{x} = Ax + Bu, \quad (8)$$

with matrices A and B defined using the jacobian at the equilibrium point. The values of A and B are shown below to a resolution of 3 significant figures, while computations are done with a more precise version. Additionally, small values less than 10^{-7} are approximated as 0.

$$A = \left. \frac{\partial \vec{f}}{\partial m} \right|_{(m_e, n_e)} = \begin{bmatrix} 0 & 10.1 & 0 & -0.746 & 0 & 1.0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1.0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1.0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1.0 & 0 & 0.00405 \\ 0 & 0 & -9.81 & 0 & -0.0774 & 0 & 1.16 & 0 & -0.694 & 0 \\ 0 & 0 & 0 & 9.81 & 0 & -0.155 & 0 & 0.746 & 0 & -10.1 \\ 0 & 0 & -0.0397 & 0 & -1.25 & 0 & -9.42 & 0 & 9.41 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3.75 & 0 & -12.0 & 0 & 0.799 \\ 0 & 0 & 0 & 0 & 0.764 & 0 & -10.4 & 0 & -5.15 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.201 & 0 & -0.393 & 0 & -0.161 \end{bmatrix}, \quad (9)$$

$$B = \left. \frac{\partial \vec{f}}{\partial n} \right|_{(m_e, n_e)} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0.22 & 0.22 \\ 0 & 0 \\ -2.94 & -2.94 \\ -34.2 & 34.2 \\ -30.7 & -30.7 \\ 0.509 & -0.509 \end{bmatrix}. \quad (10)$$

C. Controllability

Next, we can establish controllability of the linearized system. Showing controllability confirms all modes can be influenced by the inputs, ensuring a valid solution and sufficient control about the equilibrium point. The linear system (8) is controllable if and only if

$$\mathbf{W} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \cdots & \mathbf{A}^n \mathbf{B} \end{bmatrix} \quad (11)$$

is full rank. In this case, plugging in $\mathbf{A}$ and $\mathbf{B}$ and using sympy to determine rank results in the finding that $\mathbf{W}$ is full rank, and, therefore, the linearized system is controllable.

D. LQR Statement of Problem

By defining vector equations

$$g(x, u) = x^T \mathbf{Q}x + u^T \mathbf{R}u, \quad (12)$$

$$h = x^T \mathbf{M}x \quad (13)$$

allows us to write the linear quadratic regulator (LQR) minimization problem.

$$\underset{u_{[t_0, t_1]}}{\text{minimize}} \quad x(t_1)^T \mathbf{M}x(t_1) + \int_{t_0}^{t_1} \left(x(t)^T \mathbf{Q}x(t) + u(t)^T \mathbf{R}u(t) \right) dt \quad (14)$$

$$\text{subject to} \quad \dot{x} = \mathbf{A}x + \mathbf{B}u, \quad x(t_0) = x_0. \quad (15)$$

E. LQR solution

The minimizer problem can be written using the Hamilton-Jacobi-Bellman Equation * and (12):

$$0 = \frac{\partial v(t, x)}{\partial t} + \underset{u}{\text{minimum}} \left\{ \frac{\partial v(t, x)}{\partial t} (\mathbf{A}x + \mathbf{B}u) + x^T \mathbf{Q}x + u^T \mathbf{R}u \right\}, \quad v(t_1, x) = x^T \mathbf{M}x, \quad (16)$$

where

$$v(t, x) = x^T \mathbf{P}(t)x. \quad (17)$$

Rewriting the equation in this form creates a path to solve for a gain matrix $\mathbf{K}$. The matrix $\mathbf{P}$ is defined as a symmetric matrix satisfying $\mathbf{P}(t_1) = \mathbf{M}$. Taking the first time derivative of the minimum equation and isolating $\dot{\mathbf{P}}$ yields

$$\dot{\mathbf{P}} = \mathbf{PBR}^{-1} \mathbf{B}^T \mathbf{P} - \mathbf{PA} - \mathbf{A}^T \mathbf{P} - \mathbf{Q} \quad (18)$$

Plugging this value into the minimum equation (16) and solving for u , we find

$$u = -\mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}x. \quad (19)$$

From our state space model (8),

$$\mathbf{K} = \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} \quad (20)$$

To determine the value of $\mathbf{K}$, the diagonal matrices $\mathbf{Q}$ and $\mathbf{R}$ must be selected. These matrices represent the cost associated with deviation from equilibrium for state and input, respectively, so a simulation of the experiment can be utilized to select values of $\mathbf{Q}$ and $\mathbf{R}$ which result in stable convergence. The group weighted the matrices $\mathbf{Q}$ and $\mathbf{R}$ iteratively, utilizing larger $\mathbf{Q}$ values to improve pitch and roll responses in the attitude state, while the larger $\mathbf{R}$ values improved elevon motion.

The matrix $\mathbf{K}$ found from these $\mathbf{Q}$ and $\mathbf{R}$ matrices can then be evaluated for stability. The matrix

$$\mathbf{F} = \mathbf{A} - \mathbf{BK} \quad (21)$$

encapsulates the behavior over time of the closed-loop system, $\dot{x} = \mathbf{F}x$. If the real components of the eigenvalues of $\mathbf{F}$ are all negative, the system will converge to its equilibrium state, verifying its asymptotic stability.

*Optimal Controllers, Aerospace Engineering Reference Pages

III. Experimental Methods

A. Controller and Simulation Design

In this project, we tested the performance and safety of the designed Linear Quadratic Regulator(LQR) controller in guiding the flying wing to a stable and accurate landing. The goal of this was verifying that the feedback gain matrix $\mathbf{K}$ could maintain stability and control and under varying launch conditions, while also ensuring a smooth flight to the runway. The controller shall successfully land the flying wing on the runway at a rate greater than 85 percent over multiple trials, ensuring the verification of it's stability and reliability.

Equilibrium conditions for the system were first determined by minimizing the steady-state equations of motion, which resulted in a state defined equilibrium position, attitude and control deflections δ_r and δ_l for level flight. Linearizing about this point of operation allowed the group to yield the state-space matrices $\mathbf{A}$ and $\mathbf{B}$, which were then used to compute the LQR gain matrix. Additionally, the matrices $\mathbf{Q}$ and $\mathbf{R}$ were adjusted to balance the altitude stability and control effort, with larger $\mathbf{R}$ values yielding smoother and slower control responses.

B. Simulation Testing

The controller was implemented in Python using the framework template which was provided by Professor Bretl and Professor Chang. The aircraft was initialized 150 m ahead and 15 m above the landing runway with randomized initial airspeeds and small applied air disturbances. During each simulation run, the controller produced elevon values that continuously stabilized the glider. This in turn would reduce the flying wing's pitch and roll oscillations as it descended towards the runway. The resulting plots of display position (p_x, p_y, p_z), attitude angles (ψ, θ , and ϕ), and control inputs (δ_r, δ_l) were subsequently recorded and analyzed.

IV. Results

Measurements from 4000 simulated trials resulted in a success rate of 87.3%. Furthermore, we plotted results from 20 successful trials alongside 10 failed trials to determine the characteristics of successful and failed trials, and we plotted data from all failed trials of the simulation to identify systematic weaknesses in the design of the controller.

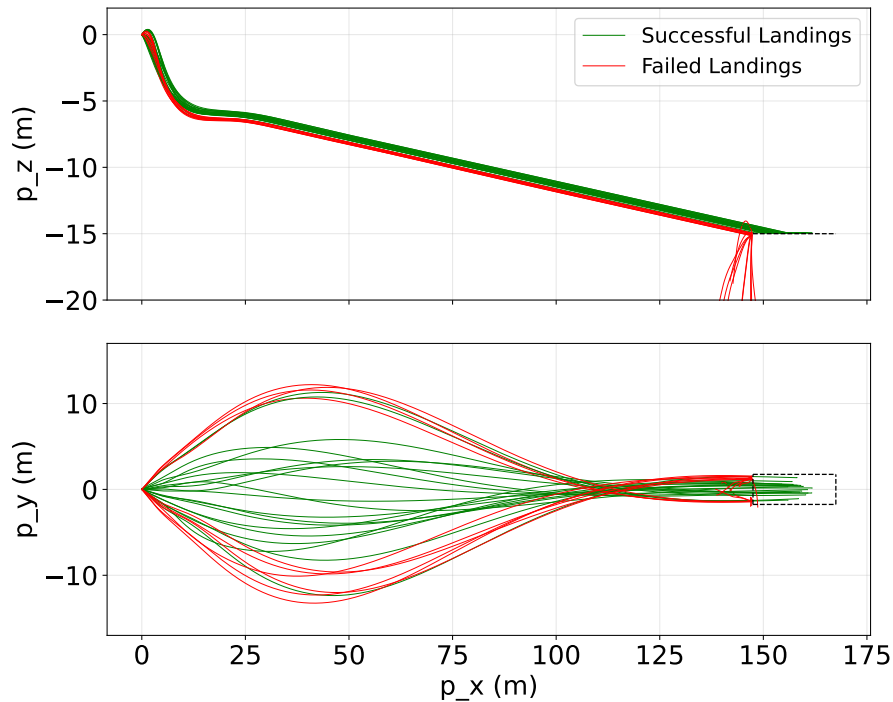


Fig. 1 These plots show the trajectory in the p_z, p_y directions over distance p_x for the flying wing for a set of successful and failed landings. Successful landings are plotted in green, while failed landings are plotted in red. A black dotted line shows the position of the landing platform.

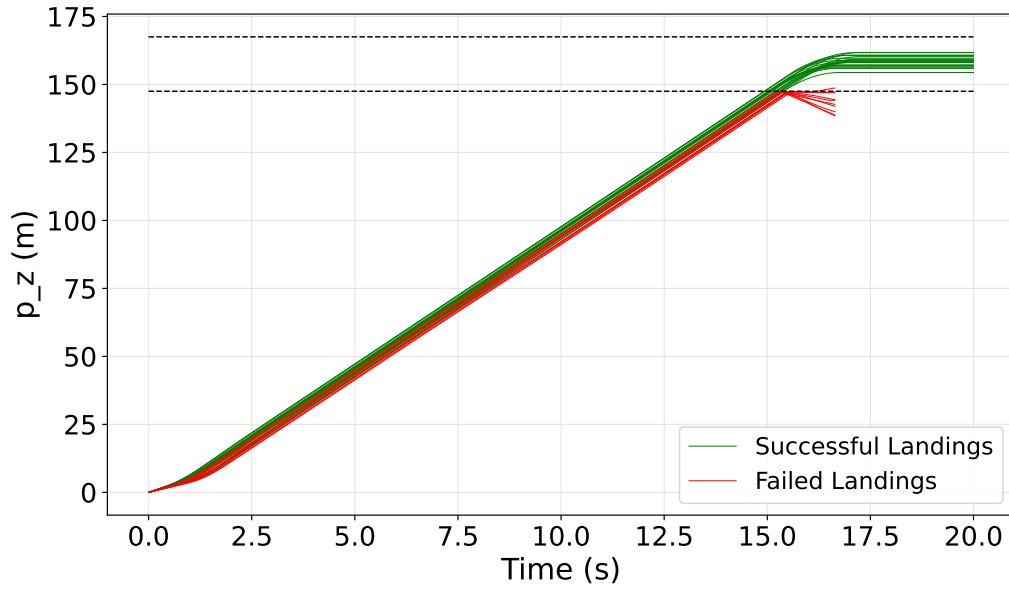


Fig. 2 This plot shows the distance p_x over time, for a set of successful and failed landings. Successful landings are plotted in green, while failed landings are plotted in red. A black dotted line shows the position of the landing platform.

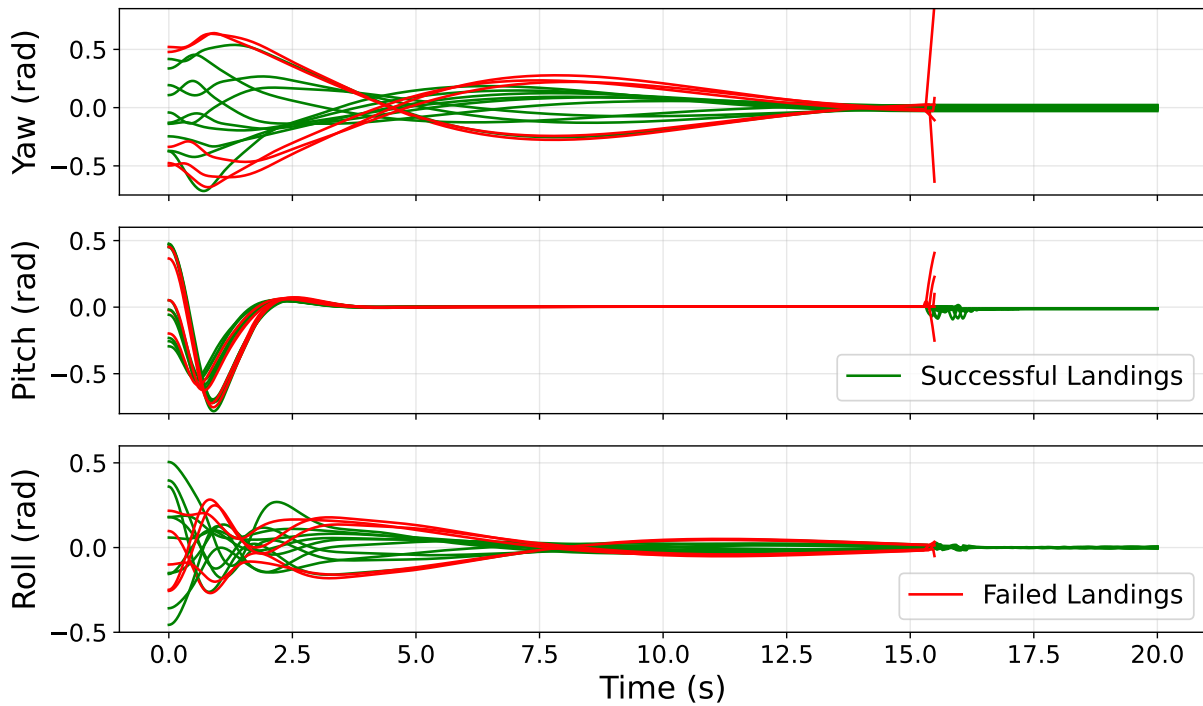


Fig. 3 These plots show pitch, roll, and yaw over time for a set of successful and failed landings. Successful landings are plotted in green, while failed landings are plotted in red.

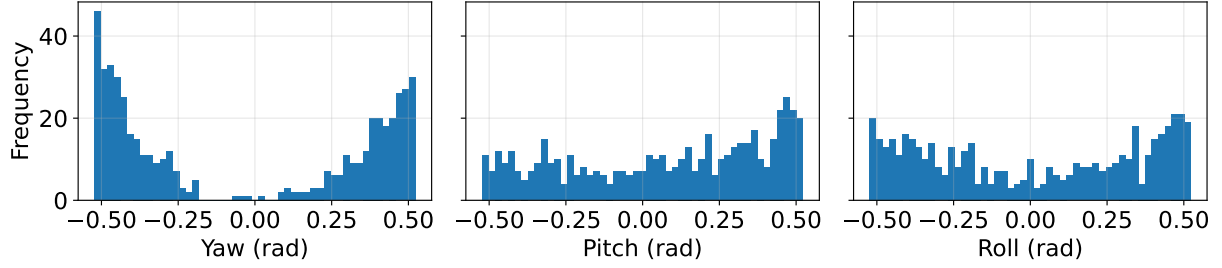


Fig. 4 These plots show the frequency of initial pitch, roll, and yaw conditions for failed landings using a histogram representation.

A. Discussion

Figures 1 and 2 show the glider’s trajectory results in the $p_x - p_y$ and $p_x - p_z$ planes. The successful landings, which are illustrated in green, were able to maintain a smooth and stable decent path with minimal path deviation, validating that the controller effectively stabilized the glider and guided it toward the runway target. The failed trajectories, which are illustrated in red, generally undershot the runway or drifted too far laterally, which goes to highlight the performance limit of the controller under extreme initial disturbances.

Figure 3 reflects the attitudes of the gliders during flight. The successful attempts of the controller were able to achieve rapid damping of yaw, pitch, and roll oscillations, converging towards zero within 15 s. This response confirms that the controller satisfies asymptotic stability, which is consistent with the LQR design’s objective. Figure 4 highlights the distribution of initial attitude deviations for failed landings. Large initial yaw and pitch offsets for the glider were reflected to be the main contributor to unsuccessful landings for the controller.

Overall, the results and data produced from the group’s code were able to validate the control design and confirm that the selected $\mathbf{Q}$ and $\mathbf{R}$ weighting matrices produced stable and well-damped behavior under the system’s conditions. Results that varied from the expected response were primarily observed when the initial orientation or yaw exceeded the target equilibrium range, which defines the effective operational limits of the controller. Therefore, the experimental results verify that the LQR controller satisfies our defined requirement for safe and stable landing performance by achieving a success landing rate of 87.3%.

V. Conclusion

This project aimed to design and test an LQR controller capable of guiding a flying wing glider to land safely on a runway with a success rate greater than 85%. First, the aircraft was linearized about a numerically determined equilibrium point, and the matrices $\mathbf{Q}$ and $\mathbf{R}$ were iteratively adjusted to achieve a stable and well damped response. Through 4000 randomized simulations, the controller achieved an 87.3% landing success rate, which validates the design requirement. The results showed effective damping of yaw, pitch, and roll oscillations as they consistently converged towards the runway. Minor performance issues would occur under large initial attitude values, which reveals the limits of the linearized controller. Overall, no significant errors were observed in our results. Aspects of the project that could be improved is that we saw that adjusting the $\mathbf{Q}$ and $\mathbf{R}$ matrices was a slow process because running simulations took a long amount of time and results had a large variation when the number of trials was few. Additionally, there seemed to be some limitations to achieving a high success rate, which we hypothesize may be connected either to the equilibrium point selection or physical limitations for an unpowered flying wing given some range of initial conditions.

VI. Acknowledgements

We thank Professor Wayne Chang and Professor Tim Bretl, along with the course assistants who aided us in understanding the project. We also thank Casey Keymolen for providing clarifications on the projects requirements.